

Jonathan Burnett

✉ jonathanburnett216@gmail.com ☎ (650) 254-6713 📍 Wimauma, FL, US

SUMMARY

Working as a Senior Data Engineer (architect, analytics) for over 10 years, I often think of data engineering like raising a child step by step. Raw data begins like a newborn — unstructured and messy. This little baby grows through data processing pipelines using various storage and stream processing technologies. As the baby grows, it becomes mature - cleaned, transformed, and structured using data processing tools within a modern data architecture. Just like guiding a child through development, data is refined through multiple layers of transformation (Bronze, Silver, Gold), reliable workflow orchestration, and scalable data processing pipelines. In its mature stage, the data becomes trusted and valuable — stored in robust data environments, powering analytics, reporting, and machine learning across the organization. Let's connect and bring up fully matured children together!

SKILLS

Languages: Python, SQL, PySpark, Scala

Data Engineering: ETL/ELT pipelines, Data Pipeline Development, Medallion Architecture, Lakehouse Architecture, Data Modeling(Star, Snowflake, Data Vault), Batch & Streaming Processing, Change Data Capture(CDC)

Big Data & Lakehouse: Apache Spark, Azure Databricks, Delta Lake, Delta Live Tables, Structured Streaming, Kafka

Cloud Platforms: Azure Data Lake Storage ADLS, Azure Databricks, Azure Data Factory, Azure Synapse Analytics, Azure Event Hubs, Azure Functions, Azure SQL, AWS S3, Glue, EMR, BigQuery

Data Storage: Azure Data Lake Storage, Amazon S3 Data Lake, Iceberg, Parquet, ORC, Avro

Orchestration: Azure Data Factory, Apache Airflow, Databricks Workflows

Databases & Warehousing: Snowflake, Azure Synapse Analytics, PostgreSQL, MySQL, Redshift

Analytics & BI: Power BI, Databricks SQL, Tableau, Looker

AI/ML Data Pipelines: MLflow, Feature Engineering, Azure Machine Learning integration, Vector Embeddings, RAG Pipelines

DevOps & Infrastructure: Terraform, Docker, Kubernetes, GitHub Actions, Jenkins, CI/CD

Governance & Security: Unity Catalog, Azure Purview, Azure RBAC, AWS Lake Formation, IAM, Data Lineage, Metadata Management, Encryption, HIPAA, SOC2

PROFESSIONAL EXPERIENCE

Deloitte & Touche LLP | Senior Data Architect/Analytics Engineer

New York, NY | Jan 2022 – current

- Led development of a Healthcare Claims & Patient Analytics Lakehouse Platform, consolidating EHR, claims, and provider datasets into a unified Azure environment using Azure Data Lake Storage and Azure Databricks, enabling population health analytics and reducing enterprise reporting latency by about 40%
- Designed the cloud data architecture on Azure using Azure Data Lake Storage as the enterprise data lake and Azure Databricks as the primary processing layer, implementing a Medallion architecture bronze, silver, gold to support scalable ingestion and analytics across multiple healthcare data sources
- Built secure data ingestion and transformation pipelines using Azure Data Factory, Azure Functions, and Databricks Auto Loader to ingest HL7, FHIR APIs, and relational healthcare systems, then

developed PySpark-based ELT and Change Data Capture CDC pipelines to standardize patient, provider, and claims datasets

- Developed batch and near real-time data pipelines using Apache Airflow, Kafka, Azure Event Hubs, and Databricks Structured Streaming, optimizing Spark workloads through partitioning strategies, caching, and Databricks autoscaling clusters to efficiently process more than 10TB of daily healthcare datasets
- Implemented governance, analytics, and ML enablement by enforcing HIPAA-compliant controls using Unity Catalog, Azure RBAC, and Microsoft Purview, deploying infrastructure with Terraform and GitHub Actions, and delivering curated datasets to business teams through Databricks SQL, Power BI, and Azure Synapse analytics, while integrating selected AWS services where required

UnitedHealth Group Inc | MLOps/Data Platform Engineer

Eden Prairie, MN | Aug 2017 - Dec 2021

- Built a Healthcare Claims Intelligence Platform consolidating claims, member eligibility, provider, and clinical datasets into an Azure-based analytics environment using Azure Data Lake Storage and Azure Databricks to support predictive healthcare analytics, fraud detection, and population health insights, improving claims risk detection and reducing manual review workloads
- Designed data ingestion and feature engineering pipelines using Azure Data Lake Storage, Azure Data Factory, and Apache Spark on Azure Databricks to process large-scale healthcare claims and clinical data, then developed Python-based feature engineering workflows to prepare datasets for machine learning models
- Developed machine learning pipelines using Python, Spark ML, and Azure Machine Learning to train and deploy predictive models for claims anomaly detection, patient readmission risk, and cost forecasting, integrating model outputs into curated analytics datasets used by healthcare operations teams
- Implemented model orchestration, monitoring, and analytics delivery using Apache Airflow and Azure Data Factory for pipeline scheduling, enforcing HIPAA-compliant security using Azure RBAC, encryption, and Key Vault, and delivering insights through Azure Synapse analytics tables and BI dashboards in Power BI and Tableau while integrating selective AWS data sources where required

Avant | Big Data/Data Pipeline Engineer

Chicago, IL | Apr 2014 - Jul 2017

- Built a Loan Risk & Customer Behavior Data Platform, developing scalable big data pipelines to process borrower applications, transaction activity, and credit bureau data using Azure Data Lake Storage and Azure Databricks, enabling faster risk analysis and improving loan decision accuracy for the fintech lending platform
- Designed the data ingestion architecture using Apache Kafka and Azure Event Hubs to capture high-volume application and transaction events, landing raw data into Azure Data Lake Storage and supporting scalable batch and streaming processing while integrating selected AWS data sources where required
- Developed distributed data processing pipelines using Apache Spark and Python on Azure Databricks to cleanse, transform, and aggregate large-scale borrower and payment datasets, producing structured analytics tables used by risk modeling and credit analytics teams
- Implemented pipeline orchestration, reliability, and analytics delivery using Apache Airflow and Azure Data Factory for workflow scheduling, optimized Spark workloads for performance and cost efficiency, and delivered curated datasets to downstream systems and BI tools through Azure Synapse Analytics and SQL-based reporting platforms

Kroger | ETL Developer / SQL Data Engineer

Cincinnati, OH | Sep 2011 - Mar 2014

- Led development of a Retail Sales & Inventory Data Platform at Kroger, integrating POS, supplier, and warehouse data into an Azure-based analytics environment using Azure Data Lake Storage and Azure Synapse Analytics, enabling near real-time reporting for merchandising, supply chain optimization, and demand forecasting across regional stores

- Designed scalable ELT pipelines using Azure Data Factory and Azure Databricks to ingest data from Oracle retail systems, flat files, and transactional POS databases, implementing staging layers, incremental loads, and data quality validations to ensure reliable processing of high-volume daily sales and inventory data
- Developed dimensional data models and optimized warehouse tables in Azure Synapse Analytics using star schema design, partitioned fact tables, and SQL-based transformations, improving query performance for business intelligence dashboards and operational reporting used by merchandising and logistics teams
- Implemented automated pipeline orchestration and monitoring using Azure Data Factory scheduling, logging frameworks, and performance tuning practices, applying error handling and data reconciliation checks that reduced pipeline failures and improved batch processing efficiency while supporting downstream reporting in Power BI

EDUCATION

University of Miami - Coral Gables, FL | B.S.Computer Science | Aug 2006 – May 2011

CERTIFICATIONS

Azure AI Engineer Associate (AI-102)
Azure Solutions Architect Expert AZ-305
Databricks Data Engineer Professional
Databricks Lakehouse Platform Architect
Microsoft Fabric Analytics Engineer Associate
Google Professional Data Engineer
Certified Kubernetes Administrator CKA

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